INDIAN INSTITUTE OF TECHNOLOGY DELHI

Department of Mathematics

Causal Inference and Machine Learning for Strategic Policy Design inHealth and Nutrition

(Based on NFHS-5)

Submitted by

Abdhesh Dash

2021MT60945

Department of Mathematics

mt6210945@iitd.ac.in

February 25, 2026

Supervisors

Rahul Singh
Department of Mathematics
Indian Institute of Technology Delhi
sirahul@iitd.ac.in

Biplab Paul
Department of Mathematics
Indian Institute of Technology Delhi
paulbiplab@maths.iitd.ac.in

Contents

1 Recap of First Semester Work **2**

1.1 Literature Review: Structural and Statistical Foundations 2

1.2 Case Study 1: Debiasing Recommender Systems via Causal Inference . . 3

1.3 Case Study 2: Causal Effect of Company Features on Stock Drawdowns During Stress Periods 3

1.4 Summary 4

2 The NFHS-5 Survey **4**

3 Causal Foundations for Structural Analysis in Population Health **5**

4 Structural Formulation of the NFHS Observational System **6**

5 Empirical Exploration and Indicator Dependency Mapping **7**

5.1 Pairwise Dependence Structure 7

5.2 Income-Stratified Correlation Analysis (SGDP-Based) 8

5.3 Area-Stratified Correlation Analysis (Rural–Urban–Total) 9

5.4 Interpretation of Stratified Dependence 10

6 Conditional Dependence Analysis and Mediation Analysis **10**

7 Graph-Based Structural Discovery and Cluster-Level Causal Modeling **13**

7.1 Causal Discovery 16

8 Future Research Directions **19**

A Appendix: Code and Notebooks **20**

A.1 Notebook A: Exploratory Data Analysis and Distribution Diagnostics . . 20

A.2 Notebook B: Correlation Analysis, Graph Clustering, and Causal Discovery 20

1 Recap of First Semester Work

The first semester of the Master's Thesis Project was devoted to building a strong theoretical and empirical foundation in causal inference, with a particular emphasis on structural learning, statistical control of false discoveries, and applications of causal machine learning in real-world systems Pearl (2009); Spirtes et al. (2000); Neal (2020). The work consisted of an in-depth literature review of three core research directions, followed by the implementation of two applied case studies that operationalized these ideas in recommendation systems and financial risk modeling.

1.1 Literature Review: Structural and Statistical Foundations

The theoretical component of the project focused on understanding how causal structures can be learned from observational data and how reliable inference can be ensured in high-dimensional settings Spirtes et al. (2000); Pearl (2009).

The first major study examined the recursive structural learning framework for Directed Acyclic Graphs (DAGs) proposed by Xie and Geng (2008). Their recursive decomposition approach provided a systematic way to factorize a joint distribution and identify causal orderings through conditional independence testing. The algorithm proceeds by iteratively identifying sink nodes and removing them, thereby reconstructing the underlying causal graph. Special attention was given to the role of conditional independence tests and the implications of replacing classical correlation measures with rank-based dependence measures. The feasibility of incorporating robust dependence statistics into recursive learning was evaluated, along with practical considerations such as sample complexity, sensitivity to noise, and computational scalability **?**.

The second line of investigation concerned the problem of controlling false discoveries in constraint-based structure learning, particularly within the PC algorithm framework Spirtes et al. (2000), as studied by Li and Wang (2009). Since DAG learning relies heavily on repeated conditional independence tests, uncontrolled error rates can lead to spurious edges or missed dependencies. The work explored how false discovery rate (FDR) control mechanisms can be integrated into the sequential testing procedure, thereby balancing structural accuracy with statistical reliability. The implications for sparse versus dense graph recovery were discussed, highlighting the trade-off between power and stability.

The third theoretical theme addressed monotonic causal effects in DAGs, following VanderWeele and Robins (2009). This framework introduced the idea of signed DAGs, where edges encode not only existence but direction of influence (positive or negative). The notions of monotonic and weakly monotonic effects were examined, along with observable inequalities that arise under such assumptions. A key insight was that violations of monotonicity constraints can serve as indicators of hidden confounding or model misspecification. Together, these three strands of literature provided a conceptual toolkit

for identifying, validating, and interpreting causal structures in observational data Pearl (2009).

1.2 Case Study 1: Debiasing Recommender Systems via Causal Inference

The first empirical case study applied causal inference principles to recommendation systems, where exposure bias and missing-not-at-random feedback pose fundamental challenges. Traditional collaborative filtering models implicitly assume that observed ratings are representative of user preferences; however, user exposure to items is itself confounded by platform policies and user behavior.

The study formulated recommendation as a causal estimation problem, where the observed rating outcome depends on both user preference and exposure mechanism. A structural framework was constructed to separate the exposure process from the preference process, enabling the estimation of deconfounded user-item effects Wang and Blei (2019). Experimental evaluation was conducted on benchmark datasets such as MovieLens Harper and Konstan (2015), Coat Becker (2012), and Yahoo! R3 Marlin (2007). Multiple estimators were compared in terms of ranking performance metrics such as NDCG@5 and Recall@5. Stability analyses were performed by artificially increasing the level of confounding, demonstrating how causal adjustments improve robustness relative to naive matrix factorization methods Athey and Imbens (2016). The case study illustrated that incorporating causal reasoning leads not only to improved predictive performance under bias but also to greater interpretability of recommendation outcomes.

1.3 Case Study 2: Causal Effect of Company Features on Stock Drawdowns During Stress Periods

The second case study investigated the causal drivers of stock drawdowns during major market stress events, specifically the COVID crash (2020) and the interest-rate hike shock (2022). While financial literature documents correlations between firm characteristics and risk, this study aimed to quantify counterfactual effects using modern causal estimators Athey and Imbens (2016).

Firm-level treatments were defined based on high price-to-earnings ratios, elevated market beta, and low profit margins. The outcome variable was maximum drawdown computed over predefined stress windows. Market capitalization and sector indicators were treated as confounders. Multiple causal estimators were implemented to estimate the Average Treatment Effect (ATE), including inverse probability weighting Austin (2011), doubly robust estimation, matching-based approaches, and generalized random forests Athey et al. (2019). Financial data were obtained using publicly available APIs such as

`yfinance` and company metadata repositories datasets organization (2020).

Beyond average effects, heterogeneous treatment effects were computed across sectors and market-cap quartiles to examine structural differences in sensitivity. Empirical results revealed that volatility (high beta) consistently exhibited the strongest negative causal effect on drawdowns across both stress regimes. Valuation effects were more regime-dependent, while profitability showed moderate but sector-sensitive influence. Flexible nonlinear estimators demonstrated greater stability across stress windows, particularly in capturing sector-level heterogeneity.

1.4 Summary

Overall, the first semester established a coherent progression from theoretical foundations in DAG learning and statistical error control to applied causal modeling in recommendation systems and financial markets. The work combined structural reasoning Pearl (2009); Spirtes et al. (2000), robust statistical inference Li and Wang (2009); Austin (2011), and modern causal machine learning Athey and Imbens (2016); Athey et al. (2019), laying the groundwork for extending these ideas further in the second phase of the thesis.

2 The NFHS-5 Survey

The empirical foundation of this study is the fifth round of the National Family Health Survey (NFHS-5), conducted during 2019–2021. NFHS-5 represents India's most comprehensive population health and demographic survey, designed to produce nationally representative estimates of fertility, nutrition, health behavior, and social determinants across all states and union territories IIPS and ICF (2021).

NFHS-5 employed a stratified, two-stage sampling design to ensure geographic and demographic representativeness. Each district was divided into rural and urban strata. Primary Sampling Units (PSUs) were selected with probability proportional to population size and a systematic sample of households was drawn from each selected PSU.

The scale of NFHS-5 is unprecedented in Indian population surveys. Approximately 636,700 households were interviewed nationwide, yielding detailed household rosters, environmental conditions, and socioeconomic profiles. Individual interviews were completed with roughly 724,000 women aged 15–49 and 102,000 men aged 15–54, covering all administrative districts in the country (about 700 districts under the NFHS-5 sampling frame). Data collection was conducted using Computer-Assisted Personal Interviewing (CAPI), which reduces transcription errors and enforces logical consistency during field entry. In addition to questionnaire modules, trained teams performed standardized anthropometric and biomarker measurements, including height, weight, haemoglobin concentration, blood pressure, blood glucose, and waist–hip circumference.

So, the NFHS-5 dataset can be viewed as a high-dimensional observational system in which socioeconomic context, health behaviours, and biological outcomes are jointly observed.

3 Causal Foundations for Structural Analysis in Population Health

Large-scale population health surveys encode rich statistical regularities, yet policy questions rarely concern association alone. Instead, they ask how outcomes would change under deliberate intervention. Distinguishing observation from intervention is therefore the conceptual starting point for structural analysis Pearl (2009); Hernán and Robins (2020). If X denotes an exposure and Y an outcome, observational summaries describe $E[Y \mid X = x]$, while causal reasoning concerns the interventional quantity

$$E[Y \mid do(X = x)],$$

which represents outcomes under active manipulation Pearl (2009). In observational systems shaped by confounding and shared latent drivers, these quantities generally differ, $E[Y \mid X = x] \neq E[Y \mid do(X = x)]$, highlighting the need for explicit structural modeling.

Causal inference formalizes intervention effects through counterfactual variables Rubin (1974); Hernán and Robins (2020). For each unit, $Y(1)$ and $Y(0)$ denote outcomes under treatment and control, and the population-level causal contrast is

$$ATE = E[Y(1) - Y(0)].$$

Identification requires assumptions linking counterfactual quantities to observed data. Conditional ignorability asserts independence of treatment and potential outcomes given covariates Z, positivity ensures sufficient overlap across exposure strata, and consistency aligns observed and counterfactual outcomes Hernán and Robins (2020). Under these principles,

$$ATE = E_Z[E[Y \mid X = 1, Z] - E[Y \mid X = 0, Z]],$$

transforming an intervention question into an estimable statistical functional.

While these ideas originate in experimental reasoning, most population health datasets are observational systems governed by interconnected social, behavioral, and biological processes. A structural representation expresses this generative mechanism as $Y = f_Y(X, Z, U_Y), X = f_X(Z, U_X)$, where unobserved disturbances encode residual variability.

Directed acyclic graphs (DAGs) provide a compact representation of such relationships, clarifying which covariates block confounding pathways and which lie on causal

chains Pearl (2009).

Machine learning enters this framework as a flexible estimation layer rather than a substitute for causal structure. High-dimensional surveys demand adaptable models for nuisance components such as conditional outcome expectations and exposure mechanisms. Chernozhukov (2018); Athey and Imbens (2019). This combination preserves causal interpretability while accommodating the complexity of real-world health data.

The key insight is that causal questions are inherently structural. Observed indicators rarely act independently; they cluster along latent socioeconomic and behavioral dimensions. Rather than treating variables as isolated predictors, the objective becomes mapping the dependency architecture that governs how exposures and outcomes co-evolve. This structural orientation sets the stage for empirical exploration and graph-based discovery in later sections.

4 Structural Formulation of the NFHS Observational System

With a causal framework established, the next step is to embed these principles within a concrete observational system. Rather than viewing the dataset as a collection of independent variables, it is treated as a structural system in which exposures, outcomes, and contextual drivers interact through shared mechanisms.

Let Y denote a health outcome of interest, X a vector of potentially modifiable indicators, and Z contextual covariates capturing socioeconomic and environmental conditions. The inferential objective concerns contrasts of the form

$$E[Y \mid do(X = x)] - E[Y \mid do(X = x')],$$

interpreted within a structural model that recognizes confounding and mediation Pearl (2009). In observational surveys, naive regression summaries $E[Y \mid X, Z]$ which describe association rather than intervention. Embedding the data within a structural causal model clarifies which dependencies arise from shared causes and which correspond to potential causal pathways.

Preparing NFHS data for structural analysis involves constructing an analytical dataset $\mathcal{D} = \{(Y_i, X_i, Z_i)\}_{i=1}^{n}$,that preserves comparability across exposure regimes. Overlap conditions, $P(X = x \mid Z = z) > 0$, ensure that intervention contrasts remain identifiable Hernán and Robins (2020), while harmonization of scales and missingness handling maintain structural coherence. These steps are not merely preprocessing tasks; they define the domain in which causal reasoning is valid.

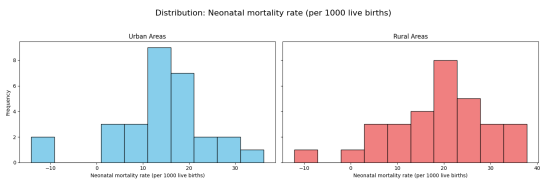
Conceptual DAGs encode hypothesized relationships among socioeconomic context,

behavioral exposures, and health outcomes. More broadly, they emphasize that NFHS indicators inhabit an interconnected system shaped by latent development gradients. Recognizing this structure motivates analytical methods capable of capturing joint dependence rather than isolated regression effects.

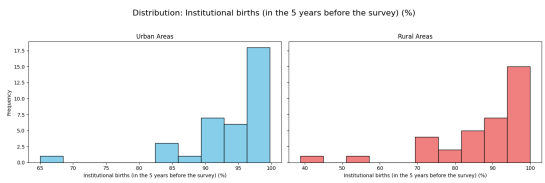
5 Empirical Exploration and Indicator Dependency Mapping

Before formal graph construction, it is necessary to examine the statistical landscape in which structural learning will operate. Empirical exploration characterizes correlation, variability, and dependence among NFHS indicators, providing intuition about the system's geometry while highlighting regions where causal assumptions may be fragile Hernán and Robins (2020).

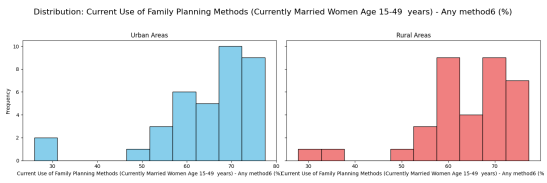
Let $(X_1, \ldots, X_p)$ denote standardized district-level indicators. Distributional analysis reveals heterogeneity across regions, a prerequisite for meaningful intervention contrasts. Indicators exhibiting limited variation signal weak support for causal comparison, whereas broad dispersion indicates structural diversity. We observe left-skewed, right-skewed, and approximately symmetric indicators, reflecting underlying demographic and socioeconomic asymmetries. Figure 1 illustrates representative distributional behaviors.



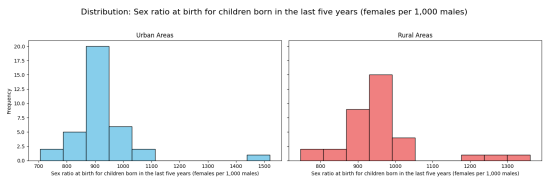
(a) Approximately Zero-Skewed Distribution (NMR)



(b) Left-Skewed Distribution (Institutional births)



(c) Left-Skewed Distribution (Family planning)



(d) Right-Skewed Distribution (Sex ratio at birth)

Figure 1: Indicator-level distribution diagnostics showing variation in skewness across development variables. The comparison highlights symmetric, left-skewed, and right-skewed behaviors that inform normalization and modeling choices.

5.1 Pairwise Dependence Structure

Pairwise dependence summarizes relational geometry via Pearson correlation:

$$\rho_{ij} = \frac{\mathrm{Cov}(X_i, X_j)}{\sqrt{\mathrm{Var}(X_i)\,\mathrm{Var}(X_j)}}.$$

Clusters of high-magnitude correlations signal latent socioeconomic coupling. From a causal perspective, such coupling implies potential confounding structures and motivates stratified analysis Pearl (2009).

	Indicator_1	Indicator_2	Correlation	Abs_Correlation	Sign
83	Female population age 6 years and above who ev...	Women (age 15-49) who are literate4 (%)	0.955962	0.955962	Positive
188	Population living in households that use an im...	Women (age 15-49) who are literate4 (%)	0.754717	0.754717	Positive
103	Population below age 15 years (%)	Population living in households with electrici...	-0.746479	0.746479	Negative
117	Population below age 15 years (%)	Women age 15 years and above wih Elevated bloo...	-0.738019	0.738019	Negative
82	Female population age 6 years and above who ev...	Population living in households that use an im...	0.675057	0.675057	Positive
189	Population living in households that use an im...	Women (age 15-49) with 10 or more years of sc...	0.658316	0.658316	Positive
115	Population below age 15 years (%)	Women age 15 years and above wih Mildly elevat...	-0.609713	0.609713	Negative
190	Population living in households that use an im...	Adolescent fertility rate for women age 15-19 ...	-0.609614	0.609614	Negative
122	Population below age 15 years (%)	Men age 15 years and above who use any kind of...	0.608350	0.608350	Positive
110	Population below age 15 years (%)	Women age 15 years and above wih very high (>1...	-0.605914	0.605914	Negative
111	Population below age 15 years (%)	Women age 15 years and above wih high or very ...	-0.579839	0.579839	Negative
84	Female population age 6 years and above who ev...	Women (age 15-49) with 10 or more years of sc...	0.575960	0.575960	Positive
107	Population below age 15 years (%)	Women (age 15-49) with 10 or more years of sc...	-0.571910	0.571910	Negative
113	Population below age 15 years (%)	Men (age 15 years and above wih very high (>1...	-0.566660	0.566660	Negative
185	Population living in households with an improv...	Men age 15 years and above who use any kind of...	-0.563874	0.563874	Negative

Figure 2: Correlation of key indicators of NFHS dataset

5.2 Income-Stratified Correlation Analysis (SGDP-Based)

Economic development acts as a global structural confounder influencing multiple indicators simultaneously. To examine whether observed dependencies persist across development regimes, states were partitioned using per-capita State Gross Domestic Product (SGDP):

$$\mathcal{S} = \mathcal{S}_{\text{low}} \cup \mathcal{S}_{\text{moderate}} \cup \mathcal{S}_{\text{high}}.$$

For each income stratum k, a correlation operator is computed, $R^{(k)} = \{\rho_{ij}^{(k)}\}$.

A mean structural correlation operator is defined as:

$$\bar{\rho}_{ij}^{\text{SGDP}} = \frac{\rho_{ij}^{(\text{low})} + \rho_{ij}^{(\text{moderate})} + \rho_{ij}^{(\text{high})}}{3}.$$

This averaging suppresses regime-specific noise while emphasizing dependencies stable across economic gradients. Persistent correlations indicate structural relationships unlikely to be artifacts of income heterogeneity.

	Indicator_1	Indicator_2	Mean_Correlation	Abs_Correlation	Sign
4860	Mothers who consumed iron folic acid for 100 d...	Mothers who consumed iron folic acid for 180 d...	0.887820	0.887820	Positive
7526	Children under 5 years who are underweight (we...	Women (age 15-49 years) whose Body Mass Index ...	0.852247	0.852247	Positive
4688	Mothers who had at least 4 antenatal care visi...	Mothers who consumed iron folic acid for 100 d...	0.842530	0.842530	Positive
6045	Children age 12-23 months fully vaccinated bas...	Children with fever or symptoms of ARI in the ...	0.820870	0.820870	Positive
505	Female population age 6 years and above who ev...	Women (age 15-49 years) having a mobile phone ...	0.807137	0.807137	Positive
5435	Institutional births (in the 5 years before th...	Home births that were conducted by skilled hea...	-0.801190	0.801190	Negative
7393	Children under 5 years who are stunted (height...	Women (age 15-49 years) who are overweight or ...	-0.797066	0.797066	Negative
6862	Children with diarrhoea in the 2 weeks precedi...	Children swith diarrhoea in the 2 weeks preced...	0.792632	0.792632	Positive
2300	Men (age 15-49) with 10 or more years of scho...	Men (age 15-49) who have ever used the intern...	0.790979	0.790979	Positive
6804	Children with diarrhoea in the 2 weeks precedi...	Children with diarrhoea in the 2 weeks precedi...	0.790895	0.790895	Positive
4689	Mothers who had at least 4 antenatal care visi...	Mothers who consumed iron folic acid for 180 d...	0.788783	0.788783	Positive
7851	Children age 6-59 months who are anaemic (<11....	All women age 15-49 years who are anaemic22 (%)	0.788644	0.788644	Positive
7849	Children age 6-59 months who are anaemic (<11....	Non-pregnant women age 15-49 years who are ana...	0.786178	0.786178	Positive

Figure 3: Income stratified correlation of key indicators of NFHS dataset

5.3 Area-Stratified Correlation Analysis (Rural–Urban–Total)

Development indicators often exhibit spatial heterogeneity. To test structural consistency across settlement contexts, correlations were computed separately for rural, urban, and aggregated populations:

$$R^{(a)} = \{\rho_{ij}^{(a)}\}, \quad a \in \{\text{rural}, \text{urban}, \text{total}\}.$$

A mean area-level dependence operator is defined as:

$$\bar{\rho}_{ij}^{\text{area}} = \frac{\rho_{ij}^{(\text{rural})} + \rho_{ij}^{(\text{urban})} + \rho_{ij}^{(\text{total})}}{3}.$$

Stable correlations across area types indicate structural coupling independent of urbanization effects, strengthening causal plausibility.

	Indicator_1	Indicator_2	Mean_Correlation	Abs_Correlation	Sign
5436	Institutional births (in the 5 years before th...	Births attended by skilled health personnel (i...	0.970078	0.970078	Positive
7241	Breastfeeding children age 6-23 months receivi...	Total children age 6-23 months receiving an ad...	0.967085	0.967085	Positive
3262	Infant mortality rate (per 1000 live births)	Under-five mortality rate (per 1000 live births)	0.957823	0.957823	Positive
399	Female population age 6 years and above who ev...	Women (age 15-49) who are literate4 (%)	0.938848	0.938848	Positive
5741	Births delivered by caesarean section (in the ...	Births in a public health facility that were d...	0.928641	0.928641	Positive
4860	Mothers who consumed iron folic acid for 100 d...	Mothers who consumed iron folic acid for 180 d...	0.915409	0.915409	Positive
5114	Mothers who received postnatal care from a doc...	Children who received postnatal care from a do...	0.914058	0.914058	Positive
5435	Institutional births (in the 5 years before th...	Home births that were conducted by skilled hea...	-0.903952	0.903952	Negative
5115	Mothers who received postnatal care from a doc...	Institutional births (in the 5 years before th...	0.836151	0.836151	Positive
7526	Children under 5 years who are underweight (we...	Women (age 15-49 years) whose Body Mass Index ...	0.831408	0.831408	Positive
3159	Neonatal mortality rate (per 1000 live births)	Infant mortality rate (per 1000 live births)	0.823721	0.823721	Positive
5589	Home births that were conducted by skilled hea...	Births attended by skilled health personnel (i...	-0.823159	0.823159	Negative
5118	Mothers who received postnatal care from a doc...	Births attended by skilled health personnel (i...	0.815757	0.815757	Positive
6238	Children age 12-23 months who have received 3 ...	Children age 24-35 months who have received a ...	0.806399	0.806399	Positive

Figure 4: Area stratified Correlation of key indicators of NFHS dataset

5.4 Interpretation of Stratified Dependence

Outcome indicators exhibit interdependence across both economic and spatial regimes, reflecting shared upstream determinants. Exposure–outcome gradients align with epidemiological intuition but remain descriptive:

$$\rho_{XY} \neq \text{causal effect.}$$

The global dependence operator,

$$R = \{\rho_{ij}\},$$

augmented by SGDP- and area-stratified operators, provides a multiscale representation of indicator coupling. High-magnitude correlations stable across strata suggest structural relationships rather than contextual artifacts.

By mapping variability across economic and spatial regimes, the NFHS dataset is revealed as a structured observational system organized along latent development axes. These insights motivate for subsequent graph-based clustering and causal discovery procedures.

6 Conditional Dependence Analysis and Mediation Analysis

Following the structural exploration of NFHS-5 indicators, the analysis progressed toward separating raw statistical association from relationships that persist under conditioning. Observational development indicators frequently co-move because they share upstream socioeconomic gradients. A central objective at this stage was therefore to determine whether strong associations represent direct structural mechanisms or are artifacts of confounding factors such as income and educational attainment.

Let X_i denote development indicators measured at the state or district level. Initial dependence was quantified using Pearson correlation, which captures linear co-movement without distinguishing causal pathways. Empirical ranking of absolute correlations revealed dense clusters linking education, digital access, and institutional health utilization.

Raw association alone, however, may be inflated by confounding variables. Conditional dependence analysis was therefore introduced through partial correlation, which removes shared variation attributable to a control set Z. For a single confounder,

$$\rho_{XY \cdot Z} = \frac{\rho_{XY} - \rho_{XZ}\rho_{YZ}}{\sqrt{(1-\rho_{XZ}^2)(1-\rho_{YZ}^2)}},$$

and more generally via regression residualization,

$$r_X = X - E[X \mid Z], \quad r_Y = Y - E[Y \mid Z],$$

$$\rho_{XY} = \mathrm{Corr}(r_X, r_Y).$$

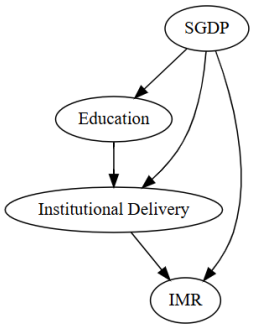
Conditioning on per-capita economic indicators and literacy attenuated many high raw correlations, revealing that a substantial portion of observed dependence arises from shared development gradients. Some associations, however, remained stable after conditioning, suggesting relationships that may warrant causal investigation. Partial correlation thus serves as a statistical approximation to confounder filtering, bridging descriptive association and structural reasoning.

	Indicator_1	Indicator_2	Partial_Correlation	Abs_Partial	Sign	Mean_Correlation_SGDP	Mean_Correlation_Area
7241	Breastfeeding children age 6-23 months receivi...	Total children age 6-23 months receiving an ad...	0.988377	0.988377	Positive	0.967085	0.936061
5436	Births attended by skilled health personnel (i...	Institutional births (in the 5 years before th...	0.984170	0.984170	Positive	0.970078	0.959897
3262	Infant mortality rate (per 1000 live births)	Under-five mortality rate (per 1000 live births)	0.984027	0.984027	Positive	0.957823	0.942223
399	Female population age 6 years and above who ev...	Women (age 15-49) who are literate4 (%)	0.957087	0.957087	Positive	0.938848	0.975438
3159	Infant mortality rate (per 1000 live births)	Neonatal mortality rate (per 1000 live births)	0.946325	0.946325	Positive	0.823721	0.959770
5741	Births delivered by caesarean section (in the ...	Births in a public health facility that were d...	0.930874	0.930874	Positive	0.928641	0.936711
3160	Neonatal mortality rate (per 1000 live births)	Under-five mortality rate (per 1000 live births)	0.922055	0.922055	Positive	0.758586	0.908272
5114	Children who received postnatal care from a do...	Mothers who received postnatal care from a doc...	0.912162	0.912162	Positive	0.914058	0.931261
7526	Children under 5 years who are underweight (we...	Women (age 15-49 years) whose Body Mass Index ...	0.889010	0.889010	Positive	0.831408	0.852247
7876	Children age 6-59 months who are anaemic (<11....	Women (age 15-49 years) having a mobile phone ...	-0.884869	0.884869	Negative	-0.409417	-0.756017
5435	Home births that were conducted by skilled hea...	Institutional births (in the 5 years before th...	-0.873376	0.873376	Negative	-0.903952	-0.801190
4688	Mothers who consumed iron folic acid for 100 d...	Mothers who had at least 4 antenatal care visi...	0.864515	0.864515	Positive	0.739032	0.842530
4860	Mothers who consumed iron folic acid for 100 d...	Mothers who consumed iron folic acid for 180 d...	0.861387	0.861387	Positive	0.915409	0.887820
7852	All women age 15-19 years who are anaemic22 (%)	Children age 6-59 months who are anaemic (<11....	0.853829	0.853829	Positive	0.731088	0.713693
5115	Institutional births (in the 5 years before th...	Mothers who received postnatal care from a doc...	0.851972	0.851972	Positive	0.836151	0.736013
4178	Children with diarrhoea in the 2 weeks precedi...	Current Use of Family Planning Methods (Curren...	-0.835003	0.835003	Negative	-0.202977	-0.266366

Figure 5: Partial correlation comparision of key indicators of NFHS dataset

Regression modeling provided an explicit framework for evaluating conditional outcome mechanisms. Ordinary Least Squares models of the form $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon$, were estimated for representative health outcomes, with education and economic indicators serving as primary controls. Results demonstrated strong negative associations between development measures and adverse health outcomes, while certain health-system utilization variables lost independent explanatory power once confounders were included. Interpreted structurally, this pattern suggests that institutional indicators may partially reflect upstream socioeconomic processes rather than acting as autonomous drivers. Below is a diagram of the proposed DAG and regression summaries.

These findings motivated construction of a preliminary Directed Acyclic Graph encoding hypothesized causal pathways linking economic conditions, educational attainment, and health outcomes. Within such a representation, parent–child relationships define conditional independence assumptions that guide model specification. Graphical abstraction ensures that statistical adjustment aligns with an interpretable causal narrative rather than arbitrary covariate inclusion.



(a) Proposed Causal DAG structure

OLS Regression Results

Dep. Variable:	Infant mortality rate (per 1000 live births)	**R-squared:**	0.618
Model:	OLS	**Adj. R-squared:**	0.588
Method:	Least Squares	**F-statistic:**	20.23
Date:	Tue, 17 Feb 2026	**Prob (F-statistic):**	5.95e-06
Time:	16:08:39	**Log-Likelihood:**	-94.092
No. Observations:	28	**AIC:**	194.2
Df Residuals:	25	**BIC:**	198.2
Df Model:	2		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P>|t|**	**[0.025**	**0.975]**
const	77.2767	11.655	6.630	0.000	53.272	101.281
Women (age 15-49) who are literate4 (%)	-0.4975	0.161	-3.100	0.005	-0.828	-0.167
Per Capita GSDP - 2020-2021	-5.214e-05	1.56e-05	-3.350	0.003	-8.42e-05	-2.01e-05

Omnibus:	0.952	**Durbin-Watson:**	1.732
Prob(Omnibus):	0.621	**Jarque-Bera (JB):**	0.472
Skew:	-0.318	**Prob(JB):**	0.790
Kurtosis:	3.023	**Cond. No.**	1.75e+06

(b) Regression Analysis results

A mediation framework further decomposed potential pathways by evaluating whether intermediate indicators transmit development effects onto health outcomes. Structural equations of the form:

$$M = \alpha_0 + \alpha_1 X + u, \quad Y = \beta_0 + \beta_1 X + \beta_2 M + \varepsilon$$

permit separation of total effects into direct and indirect components. Empirical estimation suggested that core development variables retain strong direct associations with outcomes, while mediation through institutional channels appears comparatively modest. This indicates that educational and economic influences likely operate through broad behavioral and informational mechanisms rather than solely through specific service utilization pathways.

```
###### Proportion mediated ######

beta_total = model_total.params[1]
beta_direct = model_direct.params[1]

print("Proportion mediated (in %) = ", (beta_total - beta_direct) / beta_total *100)
```

```
Proportion mediated (in %) =  -2.138572804253178
```

Figure 7: Proportion mediated (in percentage)

Taken together, conditional analysis reshapes the interpretation of NFHS indicator relationships. Strong raw correlations frequently collapse once shared confounding gradients are removed, revealing a hierarchy of structural influence centered on socioeconomic development. The analytical workflow: correlation mapping, stratified validation, partial correlation filtering, regression adjustment, graphical modelling, and mediation analysis, which forms a coherent progression from descriptive statistics to causal reasoning.

By identifying dominant confounders, isolating persistent conditional relationships,

and embedding findings within a graphical framework, the analysis prepares the ground for high-dimensional causal estimation techniques capable of modeling heterogeneous intervention effects. The NFHS dataset thus evolves from a descriptive repository into a structurally interpreted observational system, ready for rigorous causal investigation.

7 Graph-Based Structural Discovery and Cluster-Level Causal Modeling

The preceding conditional dependence analysis revealed that NFHS indicators do not behave as isolated variables; instead, they exhibit organized co-movement suggestive of latent development structures. This observation motivates a shift from pairwise association toward modular structural discovery. Rather than asking whether two indicators correlate, the analytical objective becomes identifying coherent indicator systems that represent interpretable development blocks and using these modules as scaffolding for causal reasoning. This transition reframes correlation geometry into network structure and ultimately into causal topology.

Let $X = \{X_1, \ldots, X_p\}$ denote standardized NFHS indicators aggregated at the district or state level. Pairwise association is summarized by Pearson correlation, but for structural discovery the sign of correlation is secondary to magnitude. A distance metric is therefore defined,

$$D_{ij} = 1 - |\rho_{ij}|,$$

embedding indicators in a similarity space where strong structural alignment corresponds to proximity. This transformation supports unsupervised discovery of latent modules consistent with hierarchical clustering principles Murtagh and Legendre (2014).

Agglomerative clustering minimizes inter-cluster distance,

$$d(A, B) = \frac{1}{|A||B|} \sum_{i \in A} \sum_{j \in B} D_{ij},$$

producing dendrograms that reveal nested development hierarchies. These hierarchies align with recognizable domains such as education access, maternal health, infrastructure, and socioeconomic conditions.

While hierarchical clustering emphasizes interpretability, it does not explicitly optimize network modularity, motivating a complementary graph-theoretic formulation. Indicators are therefore represented as nodes in a weighted similarity network, $G = (V, E, W)$, where edges exist whenever $|\rho_{ij}| \geq \tau$ and weights satisfy $w_{ij} = |\rho_{ij}|$. This construction

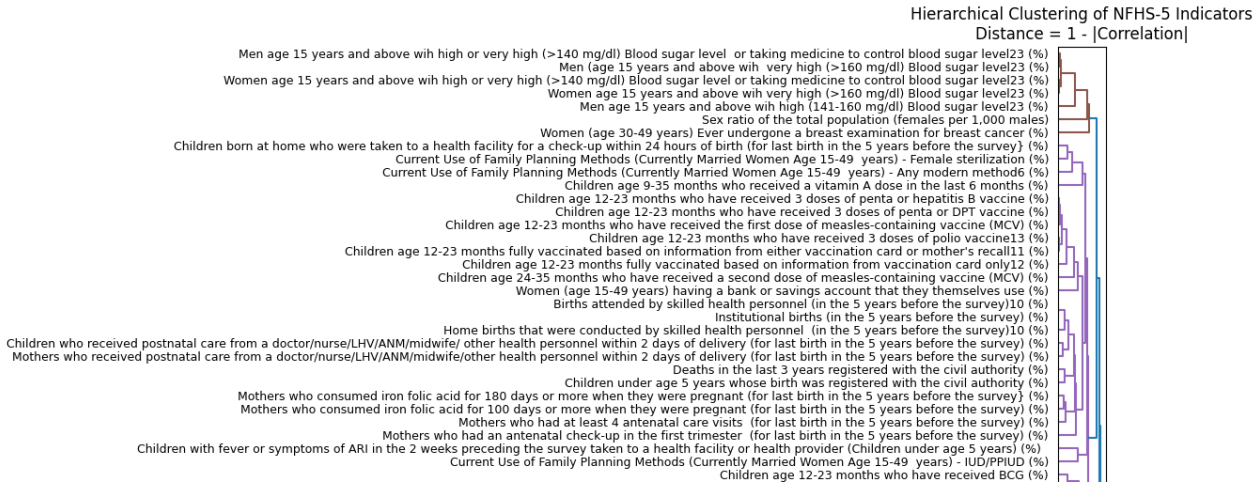


Figure 8: Hierarchical clustering dendrogram of NFHS indicators showing structural similarity based on correlation distance.

transforms correlation structure into a network whose topology encodes latent development coupling. Community detection proceeds by maximizing modularity,

$$Q = \frac{1}{2m} \sum_{ij} \left(A_{ij} - \frac{k_i k_j}{2m} \right) \delta(c_i, c_j),$$

a principle central to modern network clustering Blondel et al. (2008). Optimization via Louvain and its refinement through Leiden produces densely connected communities corresponding to interpretable development modules.

```
Louvain Module 0
 - Women (age 15-49)  with 10 or more years of schooling (%)
 - Institutional births in public facility (in the 5 years before the survey) (%)
 - Children age 12-23 months who received most of their vaccinations in a public health facility (%)
 - Children age 12-23 months who received most of their vaccinations in a private health facility (%)

Louvain Module 1
 - Population living in households that use an improved sanitation facility2 (%)
 - Children under age 3 years breastfed within one hour of birth15 (%)
 - Women (age 15-49) who are literate4 (%)
 - Female population age 6 years and above who ever attended school (%)
 - Ever-married women age 18-49 years who have experienced physical violence during any pregnancy (%)
 - Ever-married women age 18-49 years who have ever experienced spousal violence27 (%)
 - Currently married women (age 15-49 years) who usually participate in three household decisions25 (%)

Louvain Module 2
 - Children Prevalence of symptoms of acute respiratory infection (ARI) in the 2 weeks preceding the survey (Children under age 5 years) (%)
 - Prevalence of diarrhoea in the 2 weeks preceding the survey (Children under age 5 years) (%)
 - Population living in households with electricity (%)
 - Population below age 15 years (%)
 - Total Fertility Rate (number of children per woman)
 - Children under 5 years who are stunted (height-for-age)18 (%)
```

Figure 9: Louvain graph clustering on key indicators of NFHS dataset

Spectral clustering, derived from eigen-decomposition of the graph Laplacian,

$$L = D - S,$$

provides an independent geometric validation of the modular structure Ng et al. (2002). Agreement across clustering paradigms supports the robustness of latent in-

Leiden Module 0
- Children under age 5 years whose birth was registered with the civil authority (%)
- Deaths in the last 3 years registered with the civil authority (%)
- Home births that were conducted by skilled health personnel (in the 5 years before the survey)10 (%)
- Institutional births (in the 5 years before the survey) (%)
- Mothers who received postnatal care from a doctor/nurse/LHV/ANM/midwife/other health personnel within 2 days of delivery (for last birth in the 5 years before the survey) (%)
- Births attended by skilled health personnel (in the 5 years before the survey)10 (%)
- Mothers who had an antenatal check-up in the first trimester (for last birth in the 5 years before the survey) (%)
- Current Use of Family Planning Methods (Currently Married Women Age 15-49 years) - IUD/PPIUD (%)
- Children with fever or symptoms of ARI in the 2 weeks preceding the survey taken to a health facility or health provider (Children under age 5 years) (%)
- Children age 9-35 months who received a vitamin A dose in the last 6 months (%)

Leiden Module 1
- Mothers who consumed iron folic acid for 180 days or more when they were pregnant (for last birth in the 5 years before the survey} (%)
- Mothers who consumed iron folic acid for 100 days or more when they were pregnant (for last birth in the 5 years before the survey) (%)
- Births in a private health facility that were delivered by caesarean section (in the 5 years before the survey) (%)
- Men age 15 years and above wih high (141-160 mg/dl) Blood sugar level23 (%)
- Births delivered by caesarean section (in the 5 years before the survey) (%)
- Births in a public health facility that were delivered by caesarean section (in the 5 years before the survey) (%)
- Men age 15 years and above wih high or very high (>140 mg/dl) Blood sugar level or taking medicine to control blood sugar level23 (%)
- Women age 15 years and above wih high or very high (>140 mg/dl) Blood sugar level or taking medicine to control blood sugar level23 (%)
- Women age 15 years and above wih very high (>160 mg/dl) Blood sugar level23 (%)
- Men (age 15 years and above wih very high (>160 mg/dl) Blood sugar level23 (%)

Leiden Module 2
- Children age 12-23 months fully vaccinated based on information from vaccination card only12 (%)
- Women (age 15-49 years) having a bank or savings account that they themselves use (%)
- Children age 12-23 months who have received the first dose of measles-containing vaccine (MCV) (%)
- Children age 24-35 months who have received a second dose of measles-containing vaccine (MCV) (%)
- Children age 12-23 months who have received 3 doses of penta or hepatitis B vaccine (%)
- Children age 12-23 months fully vaccinated based on information from either vaccination card or mother's recall11 (%)
- Children age 12-23 months who have received 3 doses of penta or DPT vaccine (%)
- Children age 12-23 months who have received 3 doses of polio vaccine13 (%)

Figure 10: Leiden graph clustering on key indicators of NFHS dataset

Spectral Cluster 0
- Health worker ever talked to female non-users about family planning (%)
- Current users ever told about side effects of current method of family planning8 (%)

Spectral Cluster 1
- Households using clean fuel for cooking3 (%)
- Population living in households with an improved drinking-water source1 (%)
- Men (age 15-49) who have ever used the internet (%)
- Children under 5 years who are stunted (height-for-age)18 (%)
- Women (age 15-49 years) who are overweight or obese (BMI â‰¥25.0 kg/m2)21 (%)
- Men (age 15-49 years) who are overweight or obese (BMI â‰¥25.0 kg/m2) (%)
- Women age 15-24 years who use hygienic methods of protection during their menstrual period26 (%)
- Men (age 15-49 years) who know that consistent condom use can reduce the chance of getting HIV/AIDS (%)
- Women (age 15-49 years) who know that consistent condom use can reduce the chance of getting HIV/AIDS (%)
- Women (age 30-49 years) Ever undergone a screening test for cervical cancer (%)

Spectral Cluster 2
- Ever-married women age 18-49 years who have experienced physical violence during any pregnancy (%)
- Population living in households that use an improved sanitation facility2 (%)
- Ever-married women age 18-49 years who have ever experienced spousal violence27 (%)

Spectral Cluster 3
- Women (age 15-49) with 10 or more years of schooling (%)
- Women age 15-19 years who were already mothers or pregnant at the time of the survey (%)
- Adolescent fertility rate for women age 15-19 years5
- Men (age 15-49) with 10 or more years of schooling (%)
- Sex ratio at birth for children born in the last five years (females per 1,000 males)
- Mothers who consumed iron folic acid for 180 days or more when they were pregnant (for last birth in the 5 years before the survey} (%)
- Current Use of Family Planning Methods (Currently Married Women Age 15-49 years) - Pill (%)
- Women age 15 years and above who use any kind of tobacco (%)
- Women age 20-24 years married before age 18 years (%)
- Men age 15 years and above who use any kind of tobacco (%)

Figure 11: Spectral graph clustering on key indicators of NFHS dataset

dicator organization.

To facilitate higher-level reasoning, each discovered cluster C_k is summarized by a meta-variable,

$$Z_k = \frac{1}{|C_k|} \sum_{X_i \in C_k} X_i,$$

interpretable as a latent development construct. This aggregation reduces dimensional complexity while preserving shared structural information, enabling causal modeling at the module level rather than at the scale of dozens of correlated indicators.

7.1 Causal Discovery

Causal discovery is then performed within and across clusters using constraint-based graphical learning. The PC algorithm infers adjacency structure by testing conditional independencies,

$$X_i \perp X_j \mid S,$$

iteratively removing edges when independence holds and orienting remaining edges through collider rules Spirtes et al. (2000). The output is a partially directed acyclic graph (CPDAG) encoding Markov equivalence classes. In matrix form,

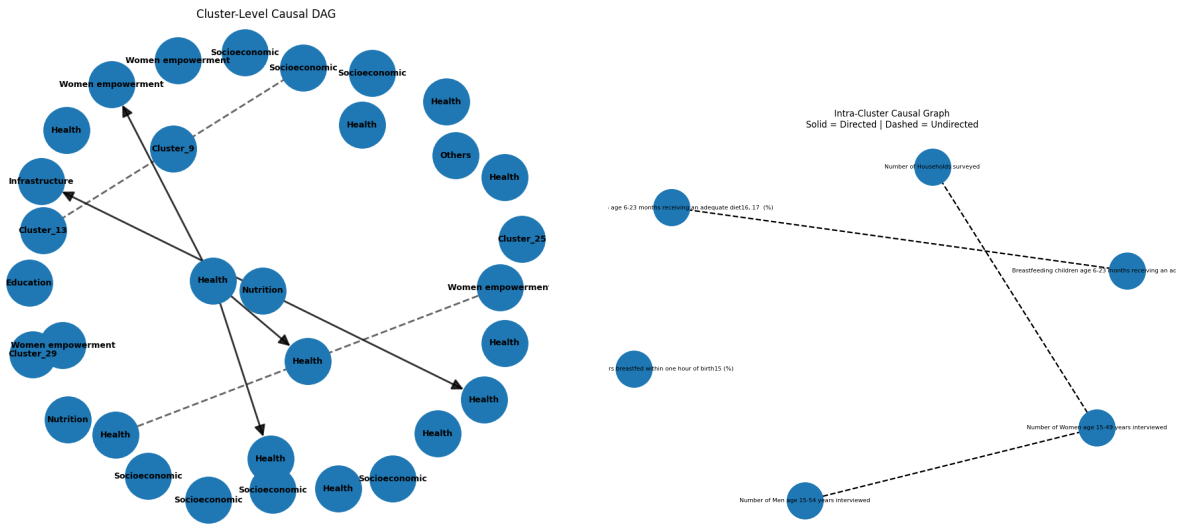
$$A_{ij} = \begin{cases} 1 & i \to j, \\ -1 & \text{undirected}, \\ 0 & \text{no edge}, \end{cases}$$

distinguishing identifiable causal directions from unresolved ambiguity.

Two structural layers emerge. Intra-cluster DAGs reveal indicator-level mechanisms operating within development domains, while inter-cluster DAGs describe higher-order causal relations between latent modules. This hierarchical representation mirrors multi-level causal reasoning in which local processes aggregate into macro-development pathways.

Directed paths in the learned graph, $X_a \to X_{i_1} \to \cdots \to X_b$, naturally correspond to mediation chains. Such pathways decompose total causal influence into direct and indirect components, aligning with structural mediation theory. Graph traversal algorithms identify candidate mechanisms linking education, healthcare access, and child outcomes, generating interpretable hypotheses for intervention analysis.

Because constraint-based discovery is sensitive to sampling variability, bootstrap aggregation is used to quantify edge stability. For bootstrap samples $\{D^{(b)}\}_{b=1}^{B}$, each yielding a graph $G^{(b)}$, edge confidence is estimated as



(a) Inter-cluster Directed Acyclic Graph (b) Intra-cluster Directed Acyclic Graph

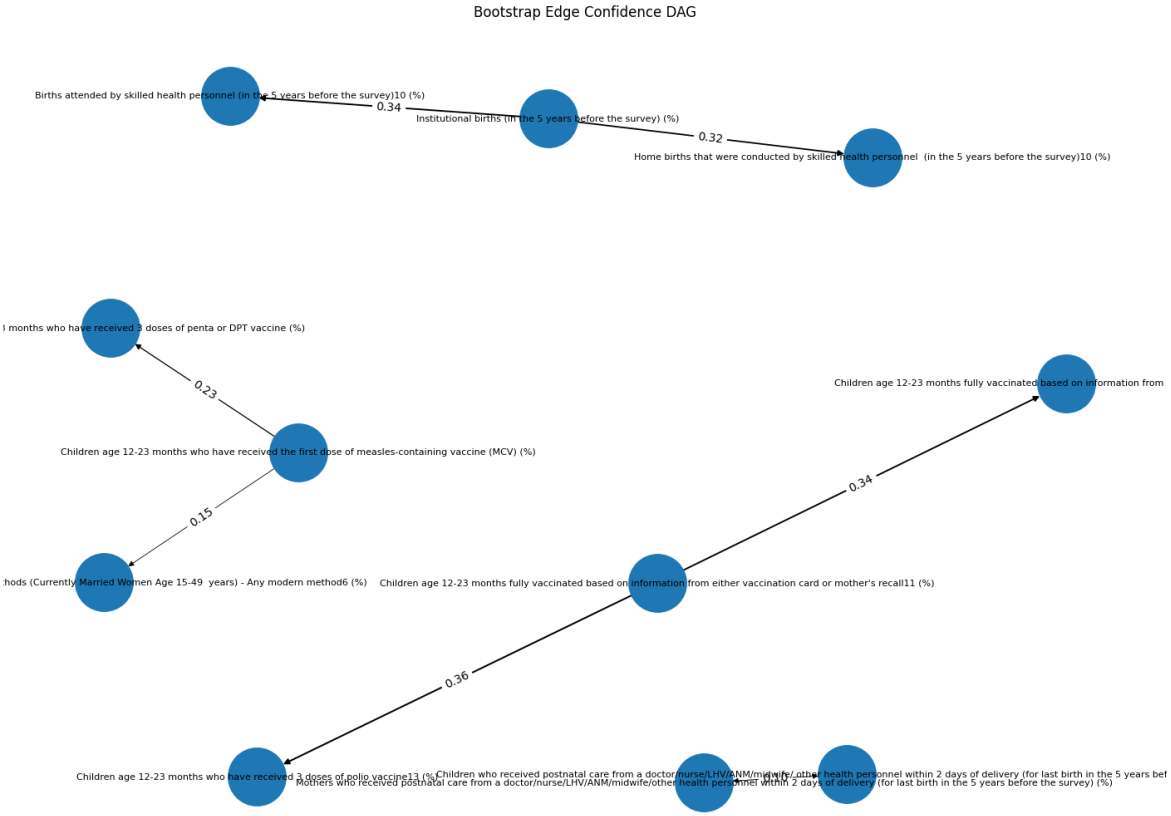
Figure 12: Comparison between macro-level (inter-cluster) and micro-level (intra-cluster) causal structures identified via graph-based clustering and causal path detection.

$$\hat{P}(i \to j) = \frac{1}{B} \sum_{b=1}^{B} \mathbf{1}\{i \to j \in G^{(b)}\}.$$

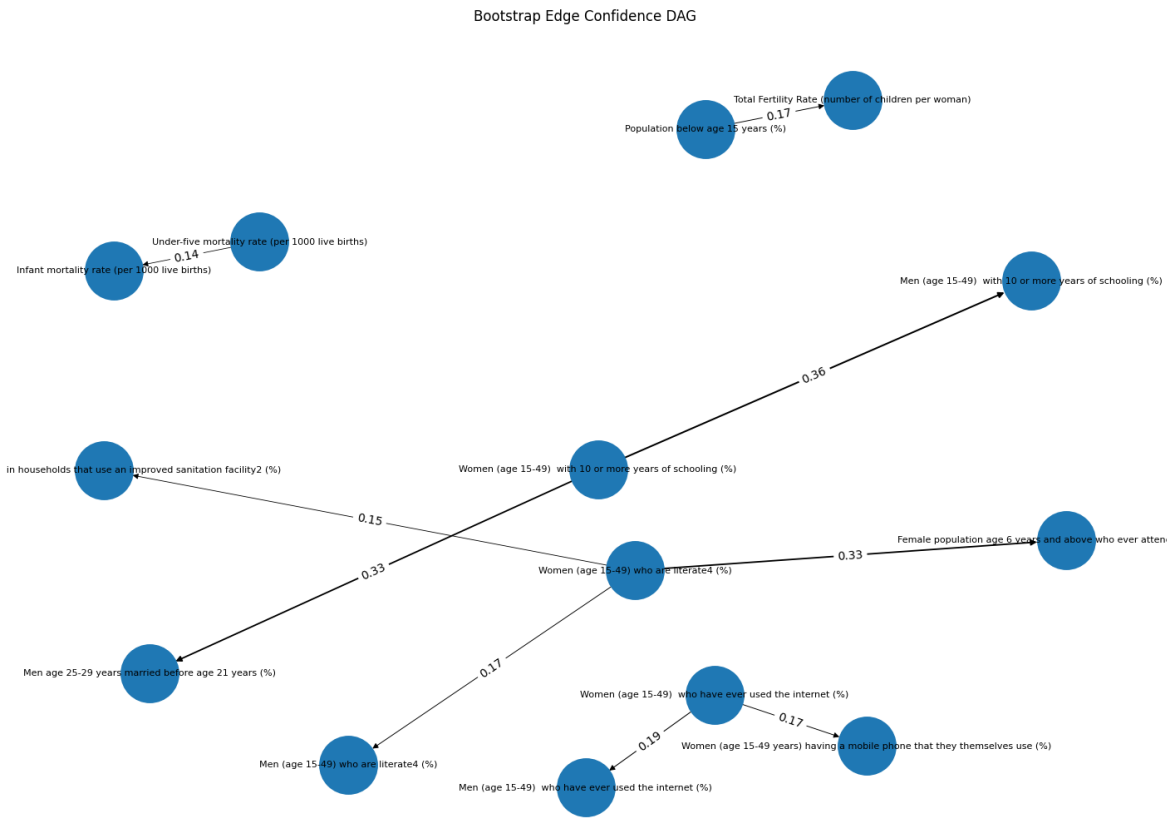
Edges only with confidence greater than a threshold value τ (here it's 0.1) are kept. High-frequency edges represent structurally stable relations, converting deterministic graph learning into probabilistic causal evidence Friedman et al. (1999). This stability analysis guards against overinterpretation of sample-specific artifacts.

Clustering quality and cross-method consistency are evaluated using modularity, silhouette structure, and agreement indices, confirming that latent modules are not artifacts of a single algorithmic choice. Structural interpretation is further enhanced through thematic filtering, whereby indicators are mapped to conceptual domains and DAG subgraphs are examined within policy-relevant contexts.

Collectively, this graph-based framework elevates NFHS analysis from correlation patterns to a hierarchical causal map of development indicators. Modular discovery identifies latent socioeconomic systems; constraint-based learning distinguishes plausible directional pathways; mediation analysis reveals intermediate mechanisms; and bootstrap aggregation provides stability-aware inference. The resulting structure forms a bridge between exploratory data geometry and formal causal modelling, establishing a scalable foundation for cluster-level intervention analysis and downstream causal machine learning estimation.



(a) Intra-cluster Directed Acyclic Graph for cluster-1



(b) Intra-cluster Directed Acyclic Graph for cluster-2

Figure 13: Intra-cluster causal structures with bootstrap edge confidence

8 Future Research Directions

The next phase of the thesis will extend the current graph-clustering-based dependency analysis toward a fully integrated structural causal learning framework. A primary objective is to connect the communities identified through Louvain, Leiden, and spectral clustering with constraint-based causal discovery algorithms, particularly the PC algorithm. The intuition is that clustering reveals modular organization in the indicator network, which can be used to restrict the search space of conditional independence tests. By limiting conditioning sets within or across structurally coherent clusters, it may be possible to reduce spurious edge detection and improve statistical efficiency. This modular strategy will be systematically evaluated in terms of false discovery rate, structural stability, and recovery of interpretable pathways.

In parallel, recursive structural learning approaches for Directed Acyclic Graphs (DAGs) will be explored as an alternative to constraint-based discovery. The recursive learning of DAGs framework provides an ordering-based decomposition of the joint distribution and may offer improved robustness in moderate-sample, high-dimensional settings. A comparative analysis between PC-based discovery and recursive learning will be conducted to assess differences in edge orientation accuracy, computational scalability, and sensitivity to noise. Particular emphasis will be placed on reducing false positive edges while preserving meaningful causal pathways.

Once validated causal DAGs are obtained, the focus will shift from structure identification to effect estimation. Identified exposure–outcome pathways will be operationalized within potential outcomes frameworks using methods such as propensity score matching, inverse probability weighting, and doubly robust estimators. In addition, modern machine learning approaches for heterogeneous treatment effect estimation will be applied to quantify variation in causal impacts across demographic and regional subgroups. This transition from structural discovery to effect estimation will enable counterfactual analysis grounded in empirically learned causal graphs.

Finally, the ultimate objective is to translate these methodological advances into socially relevant applications. The derived causal pathways will be used to inform policy-level insights in domains such as maternal and child health, nutrition, and women's empowerment. By identifying leverage points—indicators whose improvement yields significant downstream effects, the framework aims to support evidence-based intervention prioritization. The integration of graph-based structural learning, robust causal effect estimation, and policy interpretation will position the thesis toward producing actionable insights with tangible societal impact.

A Appendix: Code and Notebooks

All empirical analyses were implemented in Python using Google Colab. For transparency and reproducibility, the complete computational workflows are documented in the following notebooks.

A.1 Notebook A: Exploratory Data Analysis and Distribution Diagnostics

This notebook contains the preliminary statistical exploration of the NFHS-5 dataset. It includes data preprocessing, handling of missing values, numeric conversion of indicators, and distributional diagnostics such as histogram plots and skewness analysis.

Google Colab Link:
`https://colab.research.google.com/drive/1emGnIb3MjfVTZj9Y_7bC_WE1QgOZo_8u`

A.2 Notebook B: Correlation Analysis, Graph Clustering, and Causal Discovery

This notebook documents the core structural learning pipeline. It includes pairwise and stratified correlation analysis, mean correlation computations across SGDP and area categories, construction of similarity and distance matrices, implementation of graph clustering algorithms (Louvain, Leiden, and Spectral clustering), and causal discovery using the PC algorithm. The notebook also contains partial correlation analysis controlling for SGDP, and cluster-level graph analysis.

Google Colab Link:
`https://colab.research.google.com/drive/1NkJ3J4xEoT9i4J-h9LBUf-RYVyVFXG4d`

These notebooks collectively provide the complete computational backbone of the thesis and ensure full reproducibility of all reported empirical findings.

References

Athey, S. and Imbens, G. (2016). Machine learning methods for estimating heterogeneous causal effects. *Statistical Science*.

Athey, S. and Imbens, G. (2019). Machine learning methods economists should know about. *Annual Review of Economics*.

Athey, S., Tibshirani, J., and Wager, S. (2019). Generalized random forests. *Annals of Statistics*, 47(2):1148–1178.

Austin, P. C. (2011). An introduction to propensity score methods for reducing the effects of confounding. *Multivariate Behavioral Research*, 46(3):399–424.

Becker, B. e. a. (2012). Understanding user behavior in choice experiments (coat dataset). Cornell University.

Blondel, V. D., Guillaume, J.-L., Lambiotte, R., and Lefebvre, E. (2008). Fast unfolding of communities in large networks. *Journal of Statistical Mechanics*.

Chernozhukov, V. e. a. (2018). Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal*.

datasets organization (2020). S&p 500 companies dataset. GitHub repository.

Friedman, N., Goldszmidt, M., and Wyner, A. (1999). Learning bayesian network structure from massive datasets. *UAI*.

Harper, F. M. and Konstan, J. A. (2015). The movielens datasets: History and context. *ACM Transactions on Interactive Intelligent Systems*.

Hernán, M. A. and Robins, J. M. (2020). *Causal Inference: What If*. Chapman & Hall/CRC.

IIPS and ICF (2021). National family health survey (nfhs-5), 2019–21: India fact sheet.

Li, J. and Wang, J. (2009). Controlling the false discovery rate in the pc algorithm. In *Advances in Neural Information Processing Systems*.

Marlin, B. e. a. (2007). Collaborative filtering and the yahoo! r3 dataset. Yahoo Research.

Murtagh, F. and Legendre, P. (2014). *Ward's hierarchical clustering method*.

Neal, B. (2020). Introduction to causal inference. Lecture Notes.

Ng, A. Y., Jordan, M. I., and Weiss, Y. (2002). On spectral clustering. *Advances in Neural Information Processing Systems*.

Pearl, J. (2009). *Causality: Models, Reasoning and Inference*. Cambridge University Press, 2nd edition.

Rubin, D. B. (1974). Estimating causal effects of treatments in randomized and nonrandomized studies. *Journal of Educational Psychology*.

Spirtes, P., Glymour, C., and Scheines, R. (2000). *Causation, Prediction, and Search*. MIT Press.

VanderWeele, T. J. and Robins, J. M. (2009). Directed acyclic graphs with monotonicity. *Epidemiology*, 20(6):883–886.

Wang, Y. and Blei, D. M. (2019). The blessings of multiple causes. *Journal of the American Statistical Association*, 114(528):1574–1596.

Xie, K. and Geng, X. (2008). Recursive structural learning of directed acyclic graphs. *Journal of Machine Learning Research*.